

PRADEEPRAJ PRABHU RAJ

Chicago, Illinois, USA

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Education

University of Illinois at Chicago

Master of Science, Computer Science (GPA: 3.88/4)

Sri Venkateswara College of Engineering

Bachelor of Engineering, Computer Science

Present

Chicago, Illinois

Aug 2020 - Jul 2024

Chennai, India

Experience

Sportthon | AI/ML Developer Intern

Jun 2025 – Present

Chicago, Illinois

- Improved feature delivery speed by 40 percent by shipping a reusable AI-agent service integrating weather, wellness, and safety inputs with LLM reasoning, used across the product.
- Increased user engagement by 30 percent by designing, A/B testing, and deploying LLM applications using open-source models integrated into production systems with telemetry and experiment tracking.
- Boosted query understanding by 15 percent through targeted fine-tuning on domain datasets, adding evaluation harnesses and regression tests to prevent quality drift.
- Delivered secure, well-documented REST APIs that power AI features, integrating OpenAI and Hugging Face, with authentication, rate limiting, and structured logging for observability.
- Built a production RAG API with FastAPI using Docling OCR and layout parsing to turn scanned and complex PDFs, Word, PowerPoint, and images into clean text and table chunks, improving retrieval precision and cutting response time through layout aware chunking, hybrid search, and caching.

KaaShiv InfoTech | Software Engineer Intern

Feb 2022 - Mar 2022

Chennai, India

- Designed and developed a secure data-sharing platform using Java, incorporating advanced encryption techniques like Attribute-Based Encryption (ABE) and Proxy Re-Encryption, enhancing data integrity by 30%.
- Automated file-sharing processes using SMTP for real-time notifications, reducing manual intervention by 40%, resulting in streamlined workflows and enhanced user experience

Skills

- **Programming Languages:** Python, Java, C, C++, SQL
- **AI & Machine Learning:** PyTorch, Transformers, CNN, RNN, Generative Models, TensorFlow, Scikit-learn, OpenAI APIs, Whisper
- **LLM Finetuning & Optimization:** Axolotl, LoRA, QLoRA, BitsAndBytes, LLMCompressor, PEFT
- **AI Agentic Frameworks:** LangChain, LangGraph, CrewAI, AutoGen, Letta, Botpress, Microsoft Semantic Kernel
- **Cloud & Deployment:** AWS (S3, EC2), Apache Airflow, Google Colab, Jupyter Notebook
- **Databases & Vector Stores:** MySQL, PostgreSQL, MongoDB, Astra DB, ChromaDB, Pinecone
- **API Development & Integration:** Node.js, Postman, Webhooks
- **Data Processing & Visualization:** Pandas, Matplotlib, Seaborn, Power BI, MS Excel
- **Web Development:** HTML, CSS, JavaScript, React, Angular, Jasmine Framework
- **Developer Tools:** Git, GitHub, Visual Studio Code, Selenium, BeautifulSoup, PySpark

Projects

Automated Business Growth with Real-Time Lead Engagement using n8n | [Production URL](#)

May 2025 - Jun 2025

- Automated form-to-database pipeline by integrating Google Sheets and Gmail in n8n, reducing manual entry time by 90% through a workflow that appends or updates form responses in real time.
- Improved **customer conversion rate by 80%** as measured by successful lead follow-ups, by automating company outreach to trigger within 1 second of form submission using n8n.

Full Stack PDF RAG Application | [GitHub](#) | [Live App](#)

Mar 2025 - May 2025

- Developed a full-stack RAG system using Next.js, OpenAI, and Astra DB to enable semantic question answering over uploaded PDFs, resulting in a 3× faster information retrieval experience compared to manual search.
- Implemented PDF parsing and chunking pipeline with LangChain and text embedding models, reducing preprocessing time by 50%.
- Deployed app on Vercel with serverless APIs, ensuring reliable uptime and seamless cross-device chat.

Fine-Tuning LLaMA 3.2 for IPL Match Strategy Prediction | [GitHub](#)

Jan 2025 - Feb 2025

- Fine-tuned LLaMA 3.2 using QLoRA and PEFT on IPL match data, achieving a 90% improvement in venue-specific toss and score predictions compared to the base model.
- Built an efficient training pipeline using Hugging Face Transformers, bitsandbytes, and LoRA adapters for 4-bit quantized fine-tuning on limited GPU resources.

Crew-AI Blog Writer AI Agent | [GitHub](#)

Dec 2024 - Jan 2025

- Developed a multi-agent automation system using the CrewAI framework, coordinating specialized agents (Researcher, Planner, Writer, Editor) to collaboratively produce outputs, demonstrating practical experience in agent orchestration and task automation.

Open-Source Contributions – NVIDIA TensorRT-LLM Repository | [GitHub](#)

- Contributed to NVIDIA's TensorRT-LLM GitHub repo by getting a pull request approved, improving open-source workflow transparency, through issue reporting, branch creation, DCO compliance, CI checks, and code review